

Fundamentals of Linear Algebra for Learning Problems.

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Summer, 2025

1 Learning Objectives.

- Understand and explore the four fundamental subspaces.
 - Connect rank, nullity, and solution spaces in linear systems.
 - Analyze when linear regression solutions are unique or not.
 - Interpret how geometry of subspaces affects learning in machine learning.
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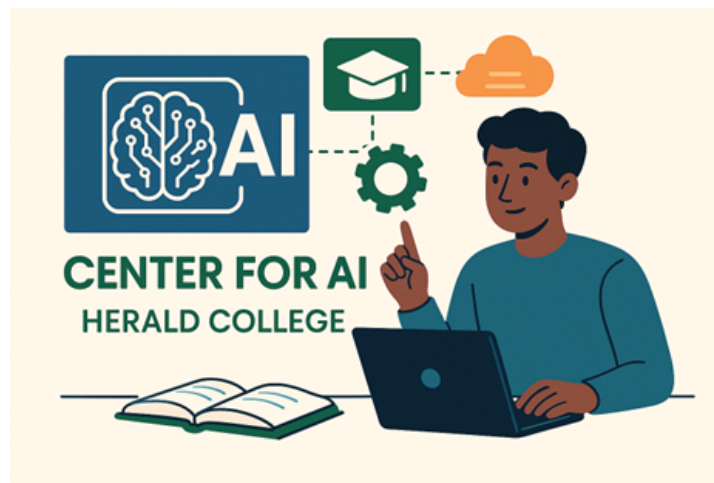


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2 Part A: Theoretical Understanding.

Definitions and Properties:

1. Define the following with examples:

- Column Space.
- Row Space.
- Null Space.
- Left Null Space.

2. Why are column rank and row rank of a matrix always equal?

- **Hint:**
 - (a) Try performing row operations to bring the matrix into row echelon form.
 - (b) What happens to the linear dependencies among the rows and columns?
 - (c) Think in terms of dimension of image space for both **A** and **A^T**.

3. If a matrix **A** has rank less than its number of columns, what can you say about its null space?

- **Hint:**
 - Recall the Rank - Nullity Theorem.

$$\text{Rank}(\mathbf{A}) + \text{Nullity}(\mathbf{A}) = \text{number of columns of } \mathbf{A}$$

- What does it mean if **nullity** > 0 ?
- Can a non - zero vector be sent to zero by **A**?

4. How does the rank of a matrix relate to the linear independence of its columns?

- **Hint:**
 - How do we define linear independence?
 - What does it mean when columns are not linearly independent?
 - How is this captured in the reduced row echelon form (RREF)?

5. If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has full column rank, what does this say about the solutions of $\mathbf{Ax} = \mathbf{b}$?

- **Hint:**
 - If columns are linearly independent, can two different **x** give the same output **Ax**?
 - Consider the geometry of the column space - Does **b** always lie in that space?
 - What does this imply for existence and uniqueness?

6. If a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has rank less than **n**, can it be invertible? Why or why not?

- **Hint:**
 - Recall the condition for invertibility: **A⁻¹exists** iff **det(A) $\neq 0$**
 - If **rank** $< \mathbf{n}$, what does that say about determinant? Can such a matrix have full row and column spaces?

3 Part B: Proof and Reasoning based Questions:

Please reason or provide a formal proof for following questions:

1. Prove the Rank - Nullity Theorem for a Matrix:

- Let $\mathbf{A}$ be an $\mathbf{m} \times \mathbf{n}$ real matrix. Consider the linear transformation:

$$\mathbf{T} : \mathbb{R}^n \rightarrow \mathbb{R}^m, \mathbf{T}(\mathbf{x}) = \mathbf{A}(\mathbf{x})$$

We define:

- The **rank** of $\mathbf{A}$, denoted as $\mathbf{rank}(\mathbf{A})$, as the dimension of the column space of $\mathbf{A}$, i.e. the number of linearly independent columns.
- The **nullity** of $\mathbf{A}$, denoted as $\mathbf{nullity}(\mathbf{A})$, as the dimension of the **null space** $\mathbf{A}$,

$$\text{i.e. the set } \mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0}$$

The **Rank - Nullity Theorem** states that:

$$\mathbf{rank}(\mathbf{A}) + \mathbf{nullity}(\mathbf{A}) = \mathbf{n}$$

- That is: the dimension of the domain of $\mathbf{T}$ (or number of columns in $\mathbf{A}$) is equal to the sum of the dimensions of the image (range) and the kernel (null space).

2. Proof Question: Injectivity and Null Space:

- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, and define the transformation $\mathbf{T}(\mathbf{x}) = \mathbf{Ax}$. Prove that $\mathbf{T}$ is **injective** if and only if $\mathbf{nullity}(\mathbf{A}) = 0$
- **Hint:**
 - What does injectivity mean in terms of solutions to $\mathbf{Ax} = \mathbf{Ay}$.
 - Try rewriting $\mathbf{Ax} = \mathbf{Ay}$ as $\mathbf{A}(\mathbf{x} - \mathbf{y}) = \mathbf{0}$. What does that say about $\mathbf{x} - \mathbf{y}$?
 - If the null space only contains $\mathbf{0}$, what can you say about the solutions?

3. Rank Nullity Application Question:

- Let $\mathbf{A} \in \mathbb{R}^{5 \times 8}$ and suppose $\mathbf{nullity}(\mathbf{A}) = 3$.
What is the **rank** of $\mathbf{A}$? What does this tell you about the **solutions to $\mathbf{Ax} = \mathbf{0}$** ?
- **Hint:**
 - Use the Rank - Nullity Theorem: $\mathbf{rank}(\mathbf{A}) + \mathbf{nullity}(\mathbf{A}) = \mathbf{n} = 8$
 - What does having nullity 3 mean in terms of number of free variables?
 - Are the columns linearly independent?

4. Contrapositive Thinking:

- If a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **not invertible** prove that **nullity**($\mathbf{A}$) > 0 .
- **Hint:**
 - Use the contrapositive of:
A matrix is invertible $\Leftrightarrow$ it has full rank $\Leftrightarrow$ **nullity** = 0.
 - Try applying **Rank - Nullity Theorem**: What must happen if rank is less than $\mathbf{n}$.

5. Reasoning: Dimension of Column Space

- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Explain why the **maximum possible rank** of $\mathbf{A}$ is **min**($\mathbf{m}$, $\mathbf{n}$), and not more.
- **Hint:**
 - How many columns can be linearly independent at most?
 - What happens when $\mathbf{m} < \mathbf{n}$? Can more than $\mathbf{m}$ columns be independent in $\mathbb{R}^{\mathbf{m}}$?

6. Short Proof: Rank Equality.

- Prove that **row rank** and **column rank** of any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ are always equal.
- **Hint:**
 - Think about Gaussian elimination and how row operations preserve row space and affect pivot positions.
 - The number of pivot positions equals both row rank and column rank after row reduction.

4 Part C: Linear Regression from a Subspace View.

Linear Regression as Projection:

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $m > n$. Consider the overdetermined system $\mathbf{Ax} \approx \mathbf{b}$.

1. Least Squares Solution and Orthogonal Projection.

- **Problem:** Show that the least squares solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

is the vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes the squared error:

$$\|\mathbf{Ax} - \mathbf{b}\|^2$$

Then explain **why** this corresponds to projecting $\mathbf{b}$ onto the **column space of $\mathbf{A}$** .

- **Hints:**

- Think of $\mathbf{Ax}$ as a **linear combination** of the columns of $\mathbf{A}$, i.e, $\mathbf{Ax} \in \text{Col}(\mathbf{A}) \subset \mathbb{R}^m$.
- The least squares problem seeks the best approximation to $\mathbf{b}$ in that subspace. That is, you are **projecting $\mathbf{b}$** onto $\text{Col}(\mathbf{A})$, denoted $\text{proj}_{\text{Col}(\mathbf{A})}(\mathbf{b})$.
- Use the condition that $\mathbf{A}\hat{\mathbf{x}}$ is the **orthogonal projection** of $\mathbf{b}$ so,

$$\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}} \perp \text{Col}(\mathbf{A})$$

- Conclude that this means $\mathbf{A}^T(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0}$, and derive:

$$\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$$

Solve this to get the normal equations and their solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

2. What if $\mathbf{A}$ is Not Full Column Rank ?

- **Problem:** What happens if $\mathbf{A}$ is not full column rank ? That is, the columns of $\mathbf{A}$ are linearly dependent.
 - Explain why $\mathbf{A}^T \mathbf{A}$ is not invertible in that case.
 - What does the **null space of $\mathbf{A}$** tell you about the solution set of the least squares problem?

- **Hints:**

- If $\mathbf{A}$ is not full column rank, then there exists a **nonzero vector $\mathbf{z} \neq \mathbf{0}$** such that $\mathbf{Az} = \mathbf{0}$. That is $\text{null}(\mathbf{A}) \neq \mathbf{0}$.
- Recall:

$$\mathbf{A}^T \mathbf{A} \mathbf{z} = \mathbf{A}^T (\mathbf{Az}) = \mathbf{A}^T \cdot \mathbf{0} = \mathbf{0}$$

so, $\mathbf{z} \in \text{null}(\mathbf{A}^T \mathbf{A}) \Rightarrow \mathbf{A}^T \mathbf{a}$ is singular (not invertible).

- Interpretation: the least squares problem still has a solution - but not a unique one. The set of minimizers is:

$$\hat{\mathbf{x}} + \mathbf{z} : \mathbf{z} \in \text{null}(\mathbf{A})$$

- The minimizer with **smallest norm** is typically chosen, and can be computed using the **Moore - Penrose pseudoinverse**:

$$\hat{\mathbf{x}} = \mathbf{A}^+ \mathbf{b}$$

5 Part D: Orthogonality in FTLA

Orthogonality in the Fundamental Theorem fo Linear Algebra (FTLA).

1. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Prove the following statements:
 - The **column space** $\mathbf{C}(\mathbf{A})$ is orthogonal to the **left null space** $\mathbf{N}(\mathbf{A}^T)$.
 - The **row space** $\mathbf{C}(\mathbf{A}^T)$ is orthogonal to the **null space** $\mathbf{N}(\mathbf{A})$.

6 Part E: Coding and Visualization.

Use Python (NumPy + Matplotlib or similar) for the following tasks.

1. Visualizing Subspaces:

- Create a 2D/3D visualization to demonstrate:
 - Column space of $\mathbf{A}$.
 - Null space direction.
- Use arrows and projection plots to illustrate data transformation.

2. Null Space Flattening:

- Construct a rank - deficient matrix $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ (with **rank2**).
- Show how vectors in null space get ” **flattened**” to zero when multiplied by $\mathbf{A}$.

3. Data Matrix Subspaces:

- Given a data $\mathbf{X} \in \mathbb{R}^{100 \times 5}$, simulate linearly dependent columns.
- Show rank deficiency and compute the null space using SVD.

————— The - End —————